Digital Discourse on Climate Security: Structural Resilience and Emotional Paradoxes during COP27

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Abstract
This study examines the architectural and emotional shifts in global climate security discussions surrounding the COP27 summit. We used a dataset of 335,112 tweets spanning June 2022 to May 2023. By integrating social network analysis with sentiment profiling, the research identifies a digital environment shaped by internal contradictions rather than simple expansion. A core finding is the central role of justice as a structural anchor. Hashtags focused on equity function as primary bridges across diverse communities. However, a structural thematic paradox emerges because the network topology appears democratic while content remains highly segregated within ideological echo chambers. This fragmentation is further shown by a dilution effect during COP27. A massive spike in tweet volume coincided with a 57.4% collapse in network density. Such data suggests that high visibility events can weaken community cohesion. The research also uncovers a resilience paradox. This shows that the Global South maintains a more hopeful and constructive tone despite facing greater climate risks. In contrast, the Global North exhibits persistent skepticism and anxiety. Furthermore, a negative correlation of -0.2161 between participation volume and emotional tone, alongside a sharp post-summit emotional decline of 14.1%, was observed. This suggests that international summits generate temporary attention that may undermine the long-term structural and emotional resilience required for effective climate advocacy.

Keywords
Climate security, COP27, social media analysis, climate justice, dilution effect, resilience paradox, structural thematic paradox, echo chambers.

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1. Introduction

Climate security has emerged as a vital arena where environmental degradation intersects with global peace and geopolitical stability. As these risks intensify, social media platforms like X (formerly Twitter) serve as real-time barometers for global opinion [1]. This platform provides a large amount of data that allows researchers to track online activity. It also helps us observe how different groups interact during environmental crises. The aim of this research is to explore both the network structure and the public sentiment regarding climate security using a combined approach.

A key feature of this study is its specific focus on time. The original dataset by Carneiro and Tucci [2] covers a long period from January 2014 to May 2023. However, this research intentionally focuses on the twelve months from June 2022 to May 2023. We selected this specific period to provide a detailed analysis of the COP27 summit held in November 2022. By including the months before and after the summit, we ensure a continuous dataset that captures the full discussion. This includes the preparation, the event itself, and the reactions that followed. This approach avoids data gaps and allows for an accurate observation of how the network and public sentiment changed over time [3].

The primary objective of this analysis is to go beyond simple volume metrics. We want to evaluate the

deeper dynamics of the discourse. Specifically, this study integrates Social Network Analysis with Social Content Analysis. We want to determine whether global climate discussions create a unified network or divide into separate groups. We compare the structural framework of the network against its sentiment patterns [4]. This research seeks to uncover hidden contradictions. For example, we look for cases where popular accounts are not actually influential. Ultimately, the analysis aims to uncover how major international events like COP27 trigger online engagement and shape the public perception of environmental security.

2. Data Source and Collection

This study uses the "Climate Security on Twitter" dataset developed by Bia Carneiro and Giulia Tucci [2]. We ideally wanted to collect data directly from the Twitter (X) API. However, recent platform restrictions and high costs make it very difficult for students to gather large amounts of past data. To ensure our project is based on accurate information, we decided to use this professionally prepared dataset. This approach allows us to maintain scientific reliability while overcoming the technical limits of the platform.

The original dataset contains tweet identifiers for approximately 636,379 unique tweets. These span from January 2014 to May 2023. Our analysis focuses on the period from June 2022 to May 2023. As mentioned in the Introduction, this timeframe captures the discourse surrounding the COP27 summit during a phase of global geopolitical crises.

The primary data collection followed a snowball sampling technique [2]. This method is particularly well suited for social media network analysis. Unlike simple keyword searching, snowball sampling captures evolving conversations. It starts with an initial set of core concepts and expands outward. This approach allows for a comprehensive view of the network as it grows around central themes.

The sampling process began using a selected set of seed queries. These were designed to reflect a global perspective. These included core concepts such as "climate security" and "planetary security." It also included conflict terms like "water conflict" and "climate fragility." The scope extended to themes of diplomacy, peacebuilding, justice, and rights. We also included multilingual keywords to represent non-English discussions. The full list of categories and keywords used in the sampling process is presented in Table 1.

Table 1: Seed Queries Used for Snowball Sampling

Category	Keywords and Hashtags
Core Concepts	#climatesecurity, #planetarysecurity, #environmentalsecurity
Conflict & Risk	#climateconflict, #waterconflict, #resourceconflict, #climatefragility
Diplomacy	#climatediplomacy, #sustainingpeace, #environmentalpeacebuilding
Justice & Rights	#climatejustice, #climatemigration, #environmentaljustice
Multilingual	#securiteclimatique (French), #seguridadclimatica (Spanish)

For the analytical phase, this study relies on three primary preprocessed components filtered to the selected timeframe.

The first component is the Annotated Content Data. This contains tweet textual information and metadata. The key variables used in this analysis are summarized in Table 2.

A key feature of this dataset is the inclusion of the variable dataset_CSTESTE$sentiment_nrc, which provides pre-calculated sentiment scores derived from the NRC Emotion Lexicon [4]. Although this lexicon is typically used to identify specific emotions like Fear or Trust, this dataset aggregates them into a single value. This value represents the overall 'polarity' of the tweet, indicating whether the content is generally positive or negative. In addition, geographic tagging through country mentions enables spatially contextualized sentiment analysis. Instead of analyzing specific emotional categories, we use this score to understand the general mood of the climate security conversation, specifically focusing on optimism versus pessimism.

Table 2: Annotated Content Data Schema

Column Name	Description	Data Type
dataset_CS$id	Unique identifier for each tweet	String (object)
hashtags_net$year	Date when the tweet was posted	String (object)
hashtags_net$hashtags	Comma-separated list of hashtags used in the tweet	String (object)
dataset_CSTESTE$ sentiment_nrc	Pre-calculated sentiment score based on NRC Emotion Lexicon	Float64
dataset_CS$ countries_mentioned	Country mentioned in tweet	String (object)

Complementing the textual data is the Network Data. This captures the structure of the climate security discourse. This component consists of two datasets: Nodes and Edges. Together they form the foundation of the co hashtag network.

The Nodes dataset represents individual hashtags used in the discourse. It includes information on their labels, frequency, and activity over time. The Edges dataset captures the relationships between hashtags. An undirected edge indicates that two hashtags appear together within the same tweet. Edge weights reflect how often these connections happen. The original structure of these datasets is summarized in Tables 3 and 4.

Table 3: Nodes Data Schema

Column Name	Description	Data Type
Id	Unique identifier for the hashtag (node)	String (object)
Label	The hashtag text	String (object)
timeset	Timestamp in Gephi dynamic interval format	String (object)
frequency	Frequency of hashtag usage	String (object)
type	Type of node (e.g., “hashtags”)	String (object)

Table 4: Edges Data Schema

Column Name	Description	Data Type
Source	Source hashtag node	String (object)
Target	Target hashtag node	String (object)
Type	Edge type (undirected)	String (object)
Id	Unique identifier for the edge	Integer (int64)
Label	Edge label (typically empty)	Float64
timeset	Timestamp in Gephi dynamic interval format	String (object)
Weight	Co-occurrence frequency weight	Integer (int64)

Together, these components facilitate the mapping of the network structure and themes. While the Nodes dataset identifies the individual actors (hashtags), the Edges dataset captures their interactions. Combined, they form a comprehensive network capable of revealing structural patterns and community clusters.

3. Data Preparation & Insights

To ensure the accuracy of this research, we established a comprehensive data processing pipeline using Python libraries such as Pandas and NetworkX.

The first phase involved loading the annotated content data, initially containing approximately 636,379 records. We ensured data integrity by converting tweet IDs to strings to prevent numerical errors. The date column was converted into standard date objects to apply a time filter between June 1, 2022, and May 31, 2023, resulting in a subset of 335,112 tweets. To ensure clarity, we simplified the column names as follows:

- dataset_CS$id to tweet_id
- hashtags_net$year to hashtags_net_date
- hashtags_net$hashtags to hashtags
- dataset_CSTESTE$sentiment_nrc to sentiment_score
- dataset_CS$countries_mentioned to country

Missing values were handled by labeling absent country tags as "Unknown" and missing hashtags as "NoHashtag." This explicitly categorizes tweets without thematic tags. To improve the accuracy of the sentiment analysis and avoid errors, we used a hashtag specific mean imputation strategy. Instead of simply filling missing scores with 0.0, we calculated the average sentiment for each specific hashtag. We used those values to fill the gaps. This ensures that the sentiment score remains true to the actual topics being discussed. Any records that could not be filled were removed to keep the dataset clean.

The network data processing for nodes and edges required a specific approach because the timestamps utilized a Gephi dynamic interval format. We wrote a custom function to convert these serial dates into readable timestamps. For the nodes data, we removed unnecessary columns and converted frequency counts to integers. We labeled missing node types as "Unknown." This resulted in 35,351 unique hashtags for the study period. The edge data was similarly filtered to retain 353,192 cooccurrence links. Connections outside the study period were dropped to ensure the graph remained relevant to the COP27 cycle. The unnecessary Label column in the edge dataset was dropped to simplify the data structure. The final profile of the processed data is summarized in Table 5. The reduction in records is a deliberate outcome of our time filtering. This ensures that the analysis is strictly focused on the discourse surrounding the COP27 cycle rather than unrelated past data.

Table 5: Processed Dataset Summary (June 2022 – May 2023)

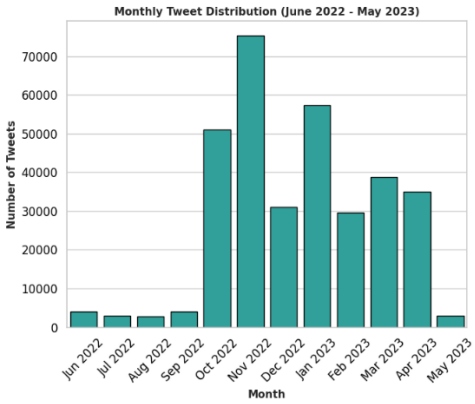
Dataset Component	Original Count	Final Count	Reduction (%)
Annotated Tweets	636,379	335,112	47.3
Network Nodes	58,019	35,351	39.1
Network Edges	547,161	353,192	35.5

The quality of this filtered dataset was validated through a thorough exploratory analysis. Our analysis of monthly tweet distribution, as shown in Fig. 1(a), confirms a highly event driven discourse. Activity remained relatively low during the summer of 2022 but experienced a massive surge in November 2022. It reached approximately 75,000 tweets. This peak matches the timeline of the COP27 summit in Sharm

El Sheikh. This proves that the dataset successfully captured the most significant climate event of the year.

Furthermore, the distribution of sentiment scores in Fig. 1(b) shows concentrations around neutral and positive values. Following our hashtag specific mean imputation, the distribution reflects a more balanced spread. The presence of multiple peaks at specific intervals indicates a rich mix of constructive policy dialogue and critical activism. This provides a robust foundation for analyzing the general sentiment polarity based on the NRC Emotion Lexicon.

(a)



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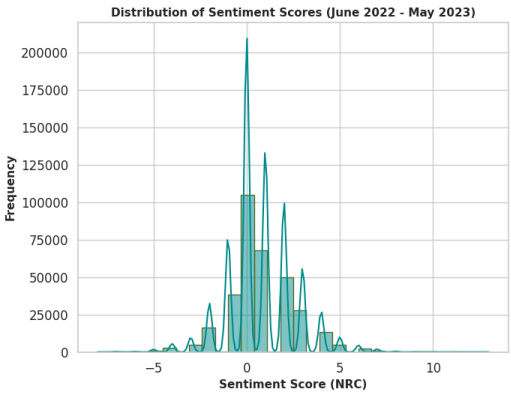
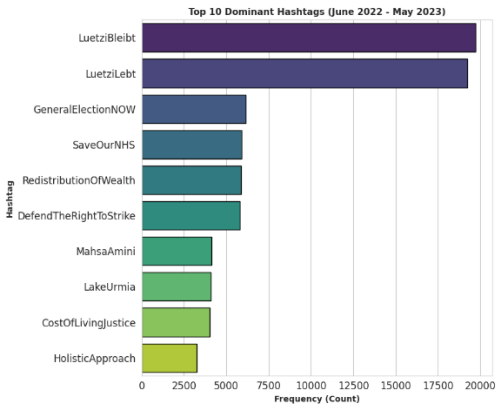


Fig. 1: Temporal and sentiment dynamics of the climate security discourse. (a) Monthly tweet distribution showing a significant surge during COP27 (November 2022). (b) Distribution of sentiment scores based on the NRC Emotion Lexicon.

The thematic landscape was explored by identifying the most dominant hashtags and their cooccurrences within the online space. As illustrated in Fig. 2(a), our analysis reveals that #LuetziBleibt was the most frequent node with 19,716 mentions. It was followed by hashtags related to broader movements like #SaveOurNHS and #GeneralElectionNOW. This suggests that during the COP27 cycle, climate security was connected with grassroots resistance movements like the Lützerath protests [5]. The structural backbone of the network is defined by the strongest cooccurrences shown in Fig. 2(b). The connection between #LuetziBleibt and #ClimateJustice emerged as the primary link with a weight of 19,563. This confirms that justice was the dominant framing mechanism used by the online community to discuss climate security. The massive cooccurrence weight between these two hashtags highlights a pivotal intersection. This is where global climate security policy met local grassroots resistance, centering "Justice" as the primary narrative bridge.

(a)



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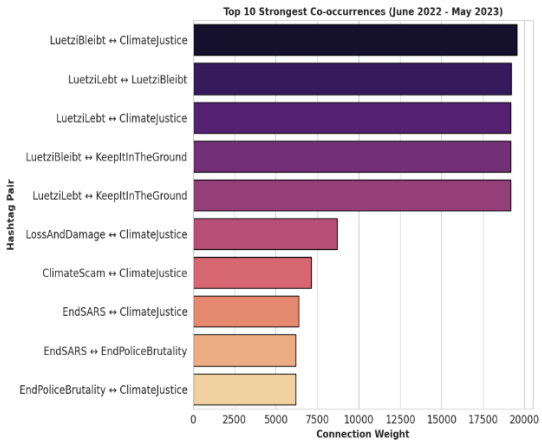

Fig. 2: Key thematic drivers and network connections. (a) Top 10 dominant hashtags, highlighting the prominence of #LuetziBleibt. (b) Top 10 strongest cooccurrence links, revealing #ClimateJustice as a central framing mechanism.

Finally, we examined the connectivity patterns of the network through the distribution of edge weights. Plotted on a log scale in Figure 3, the distribution follows a classic power law. The vast majority of connections have a weight of one while a small dense core of hashtags maintains extremely high frequency interactions. While the network is anchored by a high frequency core, the most common edge weight remains 1. This indicates a large outer edge of unique, specialized conversations that contribute to the thematic diversity of the network. This structural quality indicates a robust and resilient network where information flows through a well-defined hierarchy of

topics. This resilient core ensures that even as individual conversations fade, the overall climate security discourse remains interconnected.

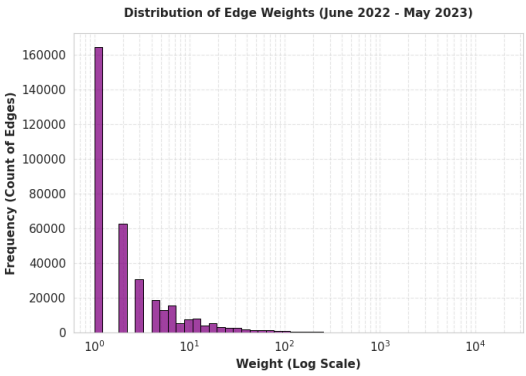

Fig. 3: Distribution of edge weights plotted on a log scale, demonstrating a power-law distribution characteristic of scale-free networks.

4. Social Network Analysis

4.1 Methodology and Metrics

The structural analysis of the cooccurrence network was conducted using Python libraries NetworkX (v2.8) and Python Louvain. We focused on the timeframe from June 2022 to May 2023. We constructed an undirected weighted graph where nodes represent individual hashtags and edges represent cooccurrence relationships. The edge weights (Weight) capture the frequency of pairwise hashtag usage within the same tweets. Upon constructing the graph from the filtered edge list, the final network comprised 41,188 unique nodes and 353,192 edges.

We employed five distinct graph theory metrics to test the study hypotheses regarding network hierarchy, thematic clustering, and influence dynamics.

First, to reveal the network hierarchy and resilience, we employed NetworkX core_number function for k core decomposition [6]. Before decomposition, we explicitly removed self-loops using nx.selfloop_edges to ensure accuracy. This process removes nodes with degree less than k at each iteration. It effectively peels away outer layers to identify the stable center or "Deep Core." This step was essential to validate the "Onion Model" of the network. It isolates the backbone of hashtags that sustain the climate security discourse.

Second, to detect natural thematic clusters, the Louvain algorithm was employed via the community_louvain.best_partition function [7]. This technique was applied to the weighted graph using the weight parameter. This ensured community detection was driven by strong ties rather than random structural associations. We set random_state=42 to ensure reproducible community assignments across runs. This choice enabled precise identification of distinct thematic sub conversations based on interaction density.

Third, to assess node level influence, we calculated two primary measures based on Freeman's framework [8]. These are Degree Centrality and Betweenness Centrality. For Betweenness Centrality, we transformed the edge weights to represent distances by creating an inv_weight attribute for each edge. Here inv_weight equals 1/Weight for edges with positive weight. This transformation was necessary because shortest path algorithms interpret weight as distance rather than connection strength. We calculated normalized betweenness centrality to obtain values between 0 and 1. Given the network scale, we employed NetworkX's approximation algorithm with a robust representative sample of k = 12,356 nodes. This is approximately 30% of the network. This ensured calculation speed while maintaining high statistical validity.

Fourth, to quantify polarization, we calculated two distinct assortativity measures [9]. First, we computed the Degree Assortativity Coefficient using NetworkX degree_assortativity_coefficient function. This metric serves as a structural test for hierarchy. Positive values indicate "Rich Club" behavior while negative values suggest a democratized structure. Second, to assess thematic isolation, we calculated the Attribute Assortativity Coefficient based on the Louvain community assignments. This metric tests for "Echo Chambers" at the content level. It determines whether hashtags within specific thematic clusters preferentially connect only with one another or mix with diverse topics.

Finally, to assess the network stability and reactivity to external shocks, we analyzed the evolution of Weighted Network Density over time [3]. Unlike standard density metrics, our custom weighted density function captures the total intensity of interactions. It calculates the ratio of the sum of edge weights to all possible connections in the graph. For each monthly subgraph, the weighted density D_w was computed using the following formula:

$$D_w = \frac{\sum w_{ij}}{N(N-1)/2}$$

where $\sum w_{ij}$ represents the sum of weights for all existing edges and N is the number of nodes in the monthly subgraph. This longitudinal approach allowed quantification of structural cohesion fluctuations. Additionally, we calculated month over month percentage growth in density to specifically

observe how major events like the COP27 summit acted as catalysts for network changes. The growth was computed as:

$$Growth = \left(\frac{D_{w,t} - D_{w,t-1}}{D_{w,t-1}}\right) \times 100\%$$

4.2 Structural Results & Discussion

Our analysis of the hashtag network from June 2022 to May 2023 reveals a strong and active network structure. The network comprises 41,188 unique nodes connected by 353,192 edges. This high connectivity suggests that the discourse is not just a collection of isolated posts. Instead, it is a tightly woven global conversation characterized by mutual engagement. The results highlight a dual structure. There is a strong central leadership driving the narrative alongside distinct thematic communities addressing specialized topics.

Network Topology: The "Onion Model"

The structural investigation, anchored by k-core decomposition, confirms a clear Core-Periphery architecture. This is often referred to as the "Onion Model" of social networks [10].

As illustrated in the shell distribution histogram (Fig. 4), the vast majority of nodes reside in the outermost shells ($k < 5$). This confirms the existence of a massive, temporary outer layer. It is composed of non-recurring participants or "one-off" hashtags that enter the discourse briefly without establishing sustained connections.

Conversely, the analysis identified a highly resilient nucleus at k = 49. This central group consists of 250 hashtags that are strongly interconnected. It includes systemic terms such as #COP27, #ClimateCrisis, and #EnvironmentalJustice. In Fig. 5, we visualized the subnetwork of hashtags with a core level greater than 15. This resulted in a dense structure of 6,874 nodes. This central group functions as the backbone of the conversation. It sustains the discourse even as activity in the outer layers diminishes.

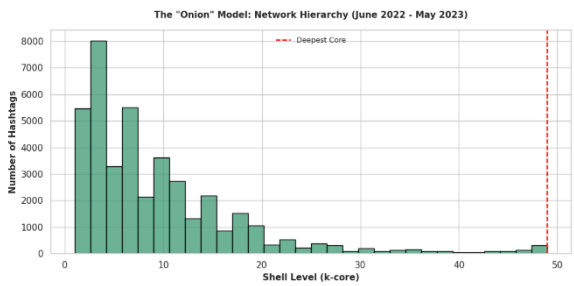

Fig. 4: Distribution of hashtags per k-core shell. The vertical red line indicates the deepest core (k = 49).

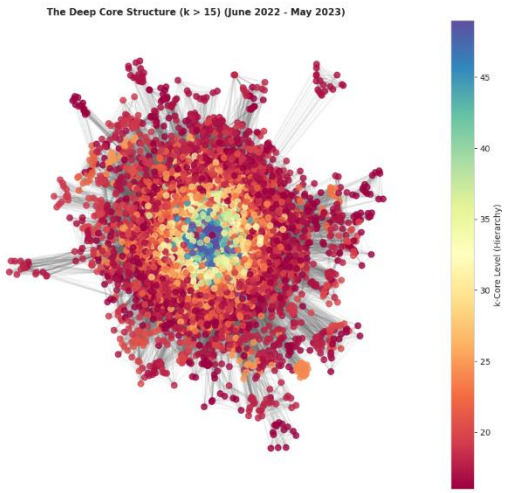

Fig. 5: Visualization of the resilient "Deep Core" structure (k > 15). Node color maps to the k-core hierarchy level using a spectral colormap. Cooler blue/purple tones represent higher k-core values (approaching the maximum of k=49) and warmer red/yellow tones indicate lower k-core values within this subset. The dense clustering demonstrates the structural cohesion of this central nucleus.

Modularity and Thematic Clusters

The application of the Louvain algorithm revealed that the climate security discourse is highly modular, with a Modularity Score (Q) of 0.5398. This relatively high score indicates a strong internal division. The network successfully partitions into 120 distinct communities. This suggests that "Climate Security" is not a single block but a composite of specialized, interconnected sub-narratives. These range from grassroots activism to geopolitical conflict.

The dominant communities representing the primary pillars of the discussion are detailed in Table 6.

Table 6: The Major Conversation Topics (Louvain Communities)

Topic	Hashtags	Main Focus	Key Examples
2	10,925	Human Rights & Justice	#MahsaAmini, #LakeUrmia, #BlackHistoryMonth, #ClimateJustice
0	8,549	Global Activism & COP27 Engagement	#WeAreNotYetDefeated, #GlobalDayofAction, #NoNewCoal, #CJPavilionCOP27
1	5,890	Finance & Skepticism	#LossAndDamages, #ClimateReparations, #ClimateScam, #DigitalCurrency
4	5,686	Protests & Resistance	#LuetziBleibt, #FreeAlaa, #BanPrivateJets, #StopEACOP
12	1,300	Civil Disobedience (XR)	#JustStopOil, #RebelForLife, #JoinTheRebellion, #CivilResistance
8	1,141	System Change & Youth	#TomorrowIsTooLate, #SystemChange, #NewGreenDeal, #YouthStrike4Climate

A detailed analysis of the keywords within these clusters is visualized in the word clouds of Fig. 6. It reveals several distinct themes. Surprisingly, the largest community (Topic 2) is not solely focused on environmental metrics. It is heavily anchored in intersectional justice. The prominence of hashtags such as #MahsaAmini and #BlackHistoryMonth alongside #ClimateJustice indicates that the public perceives climate security as part of the broader struggle for human rights. In parallel, the institutional backbone of the discourse is represented by Topic 0. This topic focuses on the COP27 summit. Keywords like #CJPavilionCOP27 and #NoNewCoal in this cluster reflect organized policy demands.

The analysis also uncovers critical insights into network polarization and resistance. Topic 1 serves as a meeting point for two opposing narratives. It includes serious discussions about financial accountability (#LossAndDamages) and skeptical ideas (#ClimateScam, #DigitalCurrency). This suggests that financial topics serve as the primary entry point for disinformation within the network. Conversely, Topic 4 is defined by tangible action. It focuses on specific locations of conflict such as the Lützerath village defense (#LuetziBleibt) and pipeline resistance (#StopEACOP). Finally, the presence of smaller communities like Topic 12 and Topic 8 represents the "rebel" wing of the network. Driven by movements like Extinction Rebellion (#RebelForLife), these groups confirm that while the main conversation focuses on policy, there is a persistent demand for faster action.

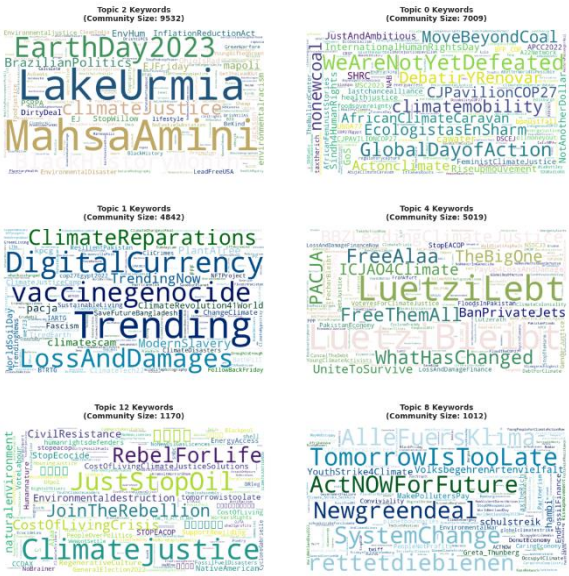

Fig. 6: Word clouds showing the main keywords for the top six dominant communities. Topic 2 (Top-Left), Topic 0 (Top-Right), Topic 1 (Middle-Left), Topic 4 (Middle-Right), Topic 12 (Bottom-Left), and Topic 8 (Bottom-Right).

Centrality Measures: The Connector Paradox

To identify the hashtags that command the greatest structural influence within the network, we calculated two primary metrics: Degree Centrality (C_D) measures local popularity. Betweenness Centrality (C_B) measures global bridging capacity or control over information flow. The analysis reveals a complex power structure where popularity does not always equal influence.

The quantitative analysis of these centrality scores is presented in Table 7. It reveals a hierarchy dominated by justice-oriented narratives but complicated by contrarian themes.

Table 7: Top 15 Bridging Hashtags (Centrality Scores).

5	Hashtag	Degree Centrality	Betweenness Centrality
1	ClimateJustice	0.4942	0.7699
2	EnvironmentalJustice	0.2488	0.2852
3	ClimateScam	0.0436	0.2741
4	climatejustice	0.2033	0.2337
5	ClimateCrisis	0.1441	0.1683
6	LossAndDamage	0.0250	0.1657
7	environmentaljustice	0.1746	0.1630
8	COP27	0.1084	0.1211
9	ClimateEmergency	0.1373	0.0848
10	climate	0.0950	0.0414
11	ClimateAction	0.1542	0.0293
12	ClimateJusticeNow	0.0492	0.0291
13	climatechange	0.0787	0.0155
14	ExtinctionRebellion	0.0133	0.0145
15	ClimateActionNow	0.0857	0.0113

To visually interpret these rankings, we mapped the relationship between popularity and influence in Fig. 7. The scatter plot reveals strategic anomalies where specific hashtags exert strong control despite limited visibility.

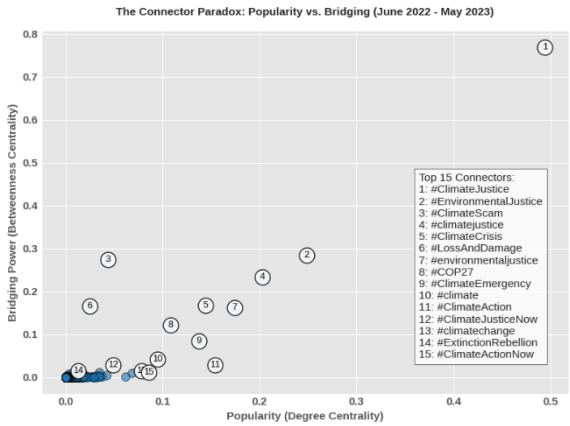

Fig. 7: The Connector Paradox: Popularity vs. Bridging. The scatter plot illustrates the relationship between Degree Centrality and Betweenness Centrality. Numbered circles highlight the top 15 bridging hashtags. It reveals anomalies where hashtags like

#ClimateScam exhibit high bridging influence despite low popularity.

The most significant finding is the dominance of #ClimateJustice. It occupies the extreme upper-right quadrant of Fig. 7. With an exceptional Betweenness score of 0.7699 and high Degree Centrality (0.494), this hashtag achieves both maximal popularity and bridging capacity. It functions as the "super-connector" of the discourse. #ClimateJustice controls the information flow across the network. This confirms that justice is not merely a topic but the primary framework of the global debate.

However, a more complex dynamic is revealed by #ClimateScam. Despite having very low popularity, it ranks third in global bridging influence. This places it in the region of "Hidden Bridges." This indicates that disinformation narratives are not isolated. Instead, they act as critical gateways linking mainstream policy discussions with skeptical communities. Similarly, #LossAndDamage exhibits high bridging power relative to its low popularity. It acts as a specialized policy bridge that connects technical debates with the broader activist network.

Conversely, hashtags like #ClimateAction exhibit high popularity ($C_D \approx 0.15$) but surprisingly low bridging power ($C_B \approx 0.03$. This demonstrates that generic activism themes often fail to act as connectors compared to specific policy topics. This highlights a paradox where influence operates through two paths: direct popularity and strategic positioning.

The analysis yields three core insights. First, the dominance of justice-oriented hashtags confirms that the justice frame serves as a universal translator between diverse communities. Second, the presence of #ClimateScam as a top bridging hashtag exposes a network vulnerability. It shows that counter-narratives can occupy influential positions. Finally, the network exhibits a disconnect between popularity and bridging power. Broad activism themes generate volume but fail to create structural cohesion.

Assortativity Analysis: Structure vs. Content

To evaluate the overall cohesion of the discourse, we performed a comparative analysis of structural and thematic mixing patterns. This approach reveals a distinction between how the network is built and how it segregates content.

Structural Integration (The Democratic Core) The analysis of Degree Assortativity yielded a negative coefficient of r = -0.1089, This confirms that the network follows a disassortative mixing pattern.

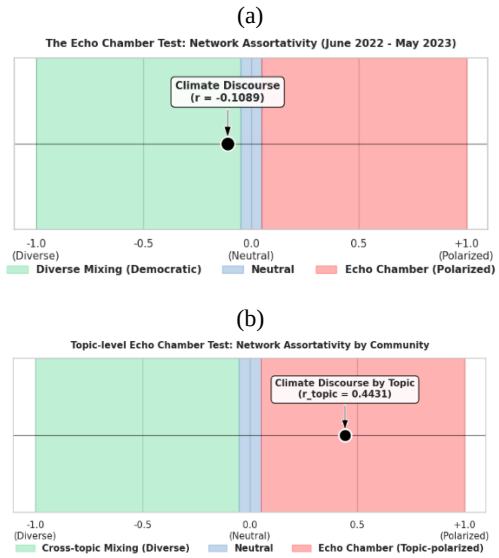

Fig. 8: Comparative Analysis of Network Mixing Patterns. (a) Structural Assortativity falls within the "Democratic" zone. (b) Topic Assortativity falls within the "Echo Chamber" zone.

This negative structural value provides critical insights. It suggests a hub-and-spoke architecture where high-degree nodes, such as #ClimateJustice, do not exclusively interact with other elites. Instead, they frequently connect with smaller, niche hashtags. This structure facilitates the efficient spread of new ideas from the outer edges to the core.

Thematic Polarization (The Content Bubbles) However, the Topic Assortativity analysis paints a contrasting picture. The positive coefficient of r_{topic} = 0.4431 indicates strong homophily at the thematic level. This means that while influencers are connecting with smaller users, the content of these conversations remains confined within specific topic clusters. For example, hashtags related to "Justice" are highly likely to co-occur with other "Justice" tags but rarely mix with "Finance" or "Skepticism" discussions.

This represents a "Structural-Thematic Paradox." The network has the potential for broad information spread due to its structure. However, it currently suffers from thematic isolation. This suggests that while the infrastructure for a global debate exists, the actual discourse is fragmented into distinct ideological groups.

Temporal Density and Dynamic Stability

To assess the resilience of the network over time, we tracked Weighted Network Density. This metric measures how tightly knit the community is. We analyzed this on a monthly basis from June 2022 to May 2023. This helps us understand whether the discourse remained a cohesive community or fragmented into a loose audience.

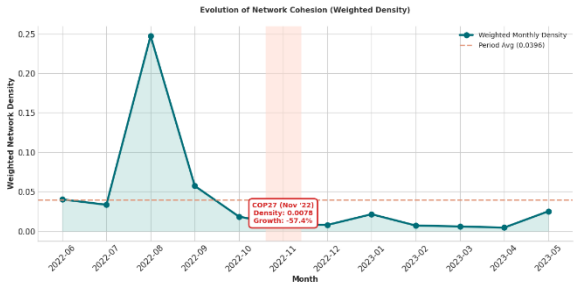

Fig. 9: Evolution of Network Cohesion (Weighted Density). The line chart tracks monthly weighted density. The red annotation highlights the COP27 summit.

The quantitative data presented in Table 8 highlights a structural shift. The discourse began with a 'High Cohesion Phase' during the summer of 2022. It reached its peak density in August 2022 (0.2474). This suggests that without a distracting mega-event, the discourse was sustained by a dedicated core of activists.

Table 8: Monthly Network Density Statistics

Months	Weighted Network Density	Structural Phase
June 2022	0.0402	Initial Cohesion
July 2022	0.0334	Steady Engagement
August 2022	0.2474	Maximum Cohesion (Activist Core)
September 2022	0.0572	High Cohesion
October 2022	0.0183	Transition Phase
November 2022	0.0078	Dilution Effect (COP27 Summit)
December 2022	0.0078	Fragmented State
January 2023	0.0214	Partial Re-engagement
February 2023	0.0070	Secondary Fragmentation
March 2023	0.0059	Fragmented State
April 2023	0.0044	Fragmented State
May 2023	0.0251	Recovery Phase

However, the approach of the COP27 summit triggered a significant structural collapse. Contrary to expectations, weighted network density dropped by 57.4% from October to November 2022. It reached its low point in November (0.0078). We define this as the "Dilution Effect." While tweet volume increased during the summit, the influx consisted largely of temporary users broadcasting messages without engaging in dialogue. This massive volume of disconnected users effectively diluted the activist core. It temporarily transformed a tight-knit community into a fragmented audience.

The network remained in a fragmented state for several months before showing recovery in May 2023. This indicates that once the temporary crowd departed, the network began to regain its structural integrity. This pattern demonstrates that while major events generate volume, they may reduce the cohesion necessary for sustained discourse.

5. Social Content Analysis

5.1 Methodology and Metrics

This phase of the research shifts the analytical focus from the network structure to the content of the messages themselves. We employ four distinct statistical models to analyze the qualitative and geographical aspects of the tweets. The foundation for this analysis is the sentiment score derived from the NRC Emotion Lexicon [4]. This score was calculated for each tweet in the dataset.

To examine regional differences and the "Resilience Paradox," we conducted Geographic Sentiment Profiling. Tweets were classified into Global North and Global South regions using a custom function applied to the country metadata field. This binary classification is a standard approach in climate justice literature regarding vulnerability [11]. Tweets with "Unknown" or missing country tags were excluded from this analysis to ensure validity. The emotional tone of each region was profiled using box plot distributions to show the spread and central tendency. We also used point estimates of the mean sentiment. To determine if the emotional difference between regions was statistically significant, we performed an Independent Samples T-test (Welch's T-test). This model statistically assesses how geographical location and socio-economic contexts influence the emotional character of digital mobilization [12].

To capture the direct emotional impact of international climate policy, we tracked Temporal Sentiment Dynamics centered on the COP27 summit. The timeline was divided into three distinct phases for comparative analysis. These are Pre-Summit (June 1 to November 5, 2022), During COP27 (November 6 to 18, 2022), and Post-Summit (November 19, 2022 to May 31, 2023). We applied a 7-day simple moving average to the daily mean sentiment scores. This calculation used a minimum period of seven days (min_periods=7) to ensure data stability. This technique smoothes out daily noise and reveals the underlying trend. This is a standard practice in social media time-series analysis [13]. We further generated a phase-based statistical summary to track how the mean sentiment changed across these strategic timeframes.

Furthermore, to measure the stability and potential polarization of the discourse, we developed a Sentiment Volatility Index. This index is defined as the daily standard deviation of sentiment scores. This

is a metric adapted from financial analytics to measure instability in public opinion [14]. We then computed a secondary 7-day rolling average on this daily volatility index. We also used a min_periods=7 threshold to maintain index integrity. This helped identify sustained periods of heightened emotional instability or rapid opinion shifts throughout the study period.

Finally, we performed a correlation analysis to investigate the Volume-Sentiment Paradox. We calculated the daily discussion volume (count of tweets) and the corresponding daily mean sentiment score. We then conducted a bivariate Spearman Rank Correlation analysis. This quantifies the strength and direction of the relationship between discourse intensity and emotional tone [15]. The resulting Spearman Rho coefficient measures the relationship. It tests whether mass mobilization is driven by collective optimism or by "negative urgency."

5.2 Content Results & Discussion

This section presents the results and discussion of the social content analysis described in Section 5.1. It examines the emotional and geographical dimensions of the climate security discourse. The analysis reveals significant emotional shifts driven by both geography and global events. The findings suggest that while the structural backbone of the network is resilient, its emotional tone is highly reactive to external shocks and regional contexts.

Regional Disparity: The Resilience Paradox

The analysis of geographic sentiment reveals a significant emotional divide between the Global North and Global South (Figure 10). The distribution of sentiment scores varies sharply across regions. The Global North exhibits both wider variance in sentiment scores and a lower central tendency compared to the Global South.

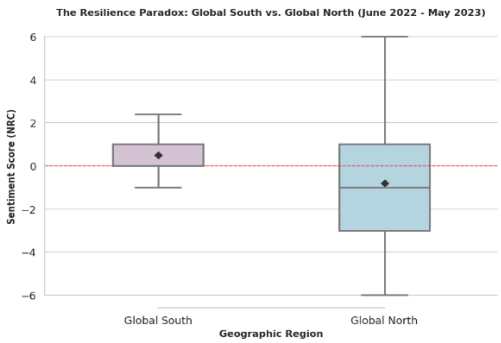

Fig. 10: The Resilience Paradox. The box plot illustrates the distribution of sentiment scores across geographic regions (June 2022 – May 2023). The black diamonds represent mean values, while the red dashed line indicates the neutral baseline.

The regional classification was performed using a custom parsing function applied to country metadata. Tweets with "Unknown" or ambiguous country tags were excluded from this specific analysis to ensure validity. The quantitative results presented in Table 9 highlight a statistical contrast between the two regions. The Global North (n=1,533) exhibits a predominantly negative mean sentiment of -0.8082, with a median score of -1.0. In contrast, the Global South (n=71,456) maintains a positive mean sentiment of 0.4854 and a median of 0.0.

Table 9: Regional Sentiment Summary (June 2022 - May 2023)

Region	Count (n)	Mean Sentiment	Std. Deviation	Median
Global North	1,533	-0.8082	2.3838	-1.0
Global South	71,456	0.4854	1.5916	0.0

The difference in mean sentiment between regions is statistically confirmed using an Independent Samples T test with Welch's correction. This accounts for unequal variances. The test yields a T-statistic of −21.1464 and an extremely significant p-value of 1.7489×10^{-87} ($p < 0.001$). This provides strong evidence against the null hypothesis of equal means. The box plot reveals distinct distribution patterns. The Global South displays a tightly clustered range with minimal outliers, suggesting emotional consensus. Meanwhile, the Global North exhibits wider dispersion (from -6 to 6) with a significant concentration of negative scores, indicating highly polarized sentiment.

This finding reveals what we define as the "Resilience Paradox." Despite possessing greater economic resources, discourse in the Global North is characterized by policy skepticism and a generally negative emotional outlook. In contrast, the Global South, despite facing higher levels of real-world climate vulnerability, exhibits a more positive profile. This suggests that in these regions, social media platforms are primarily used for constructive mobilization. They foster collective hope and solidarity rather than expressions of grievance. Furthermore, the substantial volume in the Global South reinforces the conclusion that climate security is an immediate reality for these communities, driving solution-oriented discourse.

Beyond geographical patterns, we analyzed temporal dynamics surrounding the COP27 summit.

Temporal Sentiment Dynamics: The COP27 Effect

Tracking sentiment over time confirms that international summits act as major emotional catalysts for digital communities. The discourse followed a clear three-phase evolution delineated by the COP27 summit timeline. These phases are Pre-Summit (June

1 to November 5, 2022), During COP27 (November 6 to 18, 2022), and Post-Summit (November 19, 2022 to May 31, 2023). We observed significant fluctuations throughout the study period.

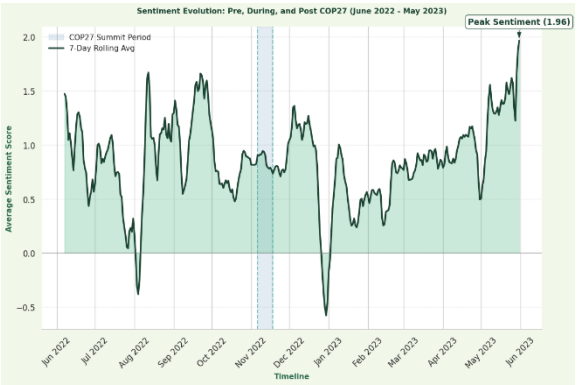

Fig. 11: Sentiment Evolution: Pre, During, and Post COP27. The dark green line represents the 7-day rolling average of sentiment scores. It reveals sustained elevation during the summit period (November 2022) and an absolute peak of 1.96 in late May 2023. Steel-blue shading marks the COP27 dates, while light green fill indicates positive sentiment periods above the neutral baseline.

The 7-day simple moving average, calculated with a minimum period of seven days (min_periods=7) ensured data stability. It effectively smoothed daily fluctuations while preserving underlying trends. The visualization highlights the COP27 summit period with a shaded background. The absolute peak sentiment value was automatically identified as 1.96.

The quantitative results presented in Table 10 provide statistical evidence of emotional shifts across phases. The discourse began with a Pre-Summit mean sentiment of 0.7653 (n=74,882). During the COP27 summit period, the mean sentiment rose to 0.8392 (n=47,360). This indicates that global conferences foster heightened states of coordinated positive engagement or trust.

Table 10: Statistical Proof of Sentiment Shift Across Phases

Phase	Count (n)	Mean Sentiment	Std. Deviation
Pre-Summit	74,882	0.7653	1.7950
During COP27	47,360	0.8392	1.6940
Post-Summit	212,870	0.7210	1.7555

The sentiment increase during COP27 represents a 9.6% rise over the Pre-Summit baseline. Notably, the standard deviation decreased during the summit period (1.6940 vs. 1.7950). This suggests greater emotional consensus or focused dialogue.

However, this summit-induced positivity proved short-lived. In the Post-Summit phase, the mean sentiment score fell to 0.7210 (n=212,870). This represents a 14.1% decline from summit levels and falls below the Pre-Summit baseline. This pattern suggests that once immediate diplomatic focus and global media attention fade, discourse often returns to a more critical or neutral baseline.

The 7-day rolling average shows a significant rebound toward the conclusion of the study period. It reached an absolute peak of 1.96 in late May 2023. This exceptional peak coincides with several climate-related developments. These include preparatory meetings for the Bonn Climate Change Conference (June 2023), increased reporting of spring 2023 extreme weather events, and growing attention to COP27 implementation. This delayed peak suggests that emotional responses to climate summits may exhibit time-lags as communities process outcomes and mobilize around policy implementation.

Visual analysis reveals distinct temporal patterns: pre-summit volatility, COP27 ascent-plateau, post-summit decline-recovery, and a late-May 2023 peak linked to specific environmental developments.

In addition to sentiment levels, we examined discourse stability through volatility analysis.

Discourse Stability and Volatility

To assess the emotional health and stability of the discourse, we developed a Sentiment Volatility Index (SVI). This index was calculated as the daily standard deviation of sentiment scores. We applied a subsequent 7-day rolling average (using min_periods=7) to smooth short-term fluctuations and reveal sustained trends in instability. This two-step methodology is adapted from financial analytics for measuring market turbulence [16]. It distinguishes isolated polarized events from prolonged contentious phases in climate security discussions.

As visualized in Figure 12, the discourse exhibited several periods of sharp volatility fluctuation. This reflects intense emotional friction or rapid opinion shifts throughout the year.

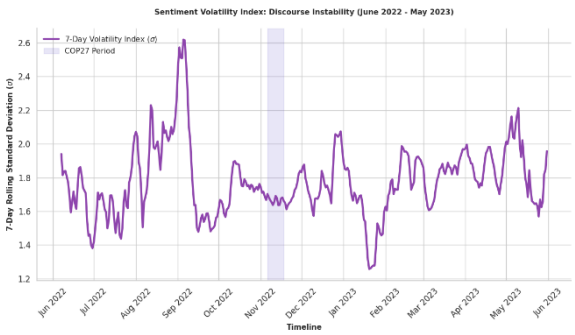

Fig. 12: Sentiment Volatility Index. The purple trend line tracks the 7-day smoothed volatility index. Higher values indicate greater discourse instability and polarization within climate security discussions.

The quantitative analysis identifies September 5, 2022, as the most volatile period. The index peaked at approximately 2.6 standard deviation units. This peak represents a dramatic fluctuation in polarization compared to the minimum volatility observed in late January 2023 (~1.25). The volatility index reveals a distinct temporal pattern: moderate instability during summer 2022 (1.6 to 2.0), peak polarization on September 5, 2022 (~2.6), gradual calming through autumn, relatively stable volatility during COP27 (~1.6), and a post-summit calming phase reaching its lowest point in late January 2023.

The September 2022 volatility peak coincides with multiple intersecting crises. These include global heatwave records, extreme weather events, the lead-up to UN General Assembly climate discussions, and heightened geopolitical tensions around energy security. This combination of factors created a perfect storm for polarized discourse. Constructive dialogue was temporarily overshadowed by intense emotional variance.

Notably, while the COP27 summit period (November 2022) saw elevated mean sentiment (0.8392), it maintained a moderate volatility of approximately 1.6 to 1.7. This is close to the study period average. This indicates that the global conference acted as an engagement catalyst with generally positive sentiment. However, it did not resolve underlying thematic tensions but rather served as a concentrated site for structured debate. The volatility index during COP27 shows initial stability followed by a slight increase toward the end of the summit. This suggests that as negotiations progressed, latent disagreements became more pronounced in the digital discourse.

The significant volatility drops to approximately 1.25 in late January 2023 represents the most emotionally stable period. This potentially reflects post-summit consensus or reduced activist mobilization during the winter months. However, the subsequent gradual increase through spring 2023 indicates that this stability was temporary. Underlying tensions resurfaced as attention shifted to implementation challenges.

This volatility analysis complements the sentiment trend analysis. It reveals that emotional consensus (low volatility) and positive sentiment do not necessarily happen together. The COP27 period demonstrated that constructive engagement can coexist with moderate polarization. Meanwhile, the post-summit period showed that declining sentiment can accompany increasing volatility. This creates particularly challenging conditions for productive climate discourse.

Finally, we investigated the relationship between discussion volume and emotional tone.

The Volume-Sentiment Paradox: Correlation Analysis

The final phase of our content analysis examines the relationship between discussion scale and emotional tone. We call this the "Volume-Sentiment Paradox."

We calculated the Spearman Rank Correlation Coefficient between daily discussion volume (count of hashtags) and daily mean sentiment scores across the study period.

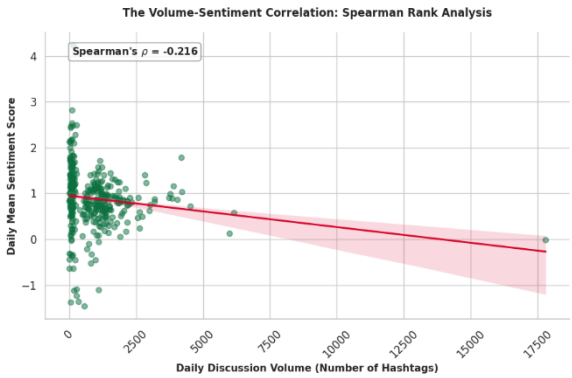

Fig. 13: The Volume-Sentiment Paradox. The regression plot illustrates the negative correlation (rho = -0.216) between daily discussion volume and mean sentiment. Dark blue data points represent individual days, while the red regression line shows the downward trend.

The correlation analysis yielded a Spearman's $\rho = -0.2161$, indicating a statistically significant negative relationship. This finding suggests that approximately 4.7% (ρ^2) of the variance in daily sentiment is linked to discussion volume. While the correlation is modest, its negative direction confirms a clear trend. As the scale of the conversation increases, the average emotional tone tends to become more critical or neutral.

Visual inspection reveals a distinct pattern. Low-volume days exhibit wide sentiment dispersion. High-volume days cluster more tightly around the regression line. The extreme outlier at approximately 17,500 daily hashtags corresponds to peak engagement periods. Here, massive participation coincided with a noticeable dip in sentiment compared to the activist-driven peaks.

This finding provides empirical support for the "Negative Urgency" hypothesis. It suggests that mass mobilization in climate security discourse is frequently catalyzed by crisis events and global summits. These periods of high visibility attract a broader, more critical audience. Their influx into the digital space effectively dilutes the concentrated positive sentiment typically maintained by the activist

core. We define this phenomenon as the "Dilution Effect." This is a process where diverse, large-scale participation moderates emotional extremes and shifts the collective discourse toward a more sober and critical baseline. The statistically significant negative correlation (-0.2161) confirms that as the volume of hashtags increases, the discourse moves away from idealistic consensus and toward a more urgent tone.

This pattern aligns with previous research indicating that digital climate activism often prioritizes urgency and critique over positive consensus-building [17]. However, it contrasts with findings from other social movements where increased participation typically correlates with rising collective efficacy. This suggests that climate security discourse follows distinct emotional dynamics characterized by "crisis-mode" mobilization.

For climate communicators, this paradox implies that strategies aimed at increasing discourse volume may inadvertently reduce sentiment positivity. Effective engagement therefore requires balancing volume expansion with mechanisms to preserve a constructive emotional tone. These observations underscore that greater visibility frequently coincides with a more critical public mood. This presents both challenges and opportunities for fostering productive and resilient climate discourse at scale.

6. Network and Sentiment Integration

To fully understand the climate security conversation, we integrated structural community data from the Social Network Analysis with emotional sentiment scores from the Social Content Analysis. This combination allows us to evaluate both the organizational patterns and the emotional drivers within the discourse. By mapping sentiment across thematic clusters and influential nodes, we identify which network parts encourage constructive dialogue and which ones show urgency or skepticism.

Community-Level Sentiment Synthesis

We merged the Louvain community assignments (from Section 4) with individual tweet sentiment scores. Hashtags were processed by lowercasing, removing "#" symbols, and splitting them to create a standard format. The resulting detailed dataset was joined with community IDs. Then we calculated the mean sentiment for each community. The top 15 communities by hashtag volume were selected for visualization. The bubble size represents relative volume. A horizontal lollipop chart maps the sentiment position relative to the global average of 0.59.

The integration of sentiment data across the top communities reveals a highly diverse emotional landscape.

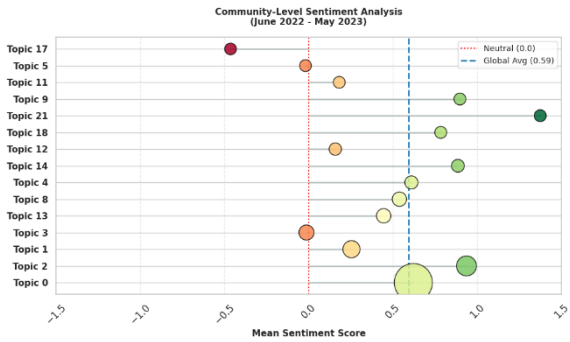

Fig. 14: Community Level Sentiment Analysis. The horizontal lollipop chart maps the mean sentiment of the top 15 communities by hashtag volume. The size of each bubble corresponds to the volume of discussion. It reveals that massive communities like Topic 0 maintain positive sentiment (0.6210). However, several high-volume clusters like Topic 3 (-0.0129) and Topic 17 trend towards neutrality or negativity.

Sentiment scores vary significantly between clusters. Topic 21 (1.3736) and Topic 2 (0.9361) exhibit the most positive tones. This indicates that certain specialized and justice-oriented discussions are framed with high optimism or resilience. In contrast, Topic 17 stands as a significant negative outlier (-0.4631). This represents a cluster likely focused on grievances, crises, or critical opposition.

Table 11: Synthesis Summary of Major Conversation Topics (Top 15 by Volume)

Topic ID	Mean Sentiment Score	Hashtag Volume	Tone Classification
21	1.3736	5,007	Highly Positive
2	0.9361	126,194	Highly Positive
9	0.8972	8,474	Highly Positive
14	0.8852	15,290	Highly Positive
18	0.7836	5,187	Highly Positive
0	0.6210	626,942	Highly Positive
4	0.6094	18,951	Highly Positive
8	0.5384	44,348	Highly Positive
13	0.4448	35,698	Positive
1	0.2539	76,298	Positive
11	0.1815	13,373	Positive
12	0.1581	11,045	Positive
3	-0.0129	45,567	Moderate/Negative
5	-0.0189	7,571	Moderate/Negative
17	-0.4631	6,263	Moderate/Negative

This distribution reveals three distinct discourse types within the climate security network. The first consists of Positive-Niche Communities (e.g., Topics 21, 14). These are small but highly constructive clusters often focused on specific solutions or solidarity. In contrast, Positive-Scale Communities (e.g., Topics 2, 0)

emerge as massive clusters. They successfully combine high discussion volume with a strongly positive tone. They effectively form the backbone of the activist movement. Finally, the analysis identifies Critical-Mass Communities (e.g., Topics 3, 17). Here significant volume coincides with negative or neutral sentiment. This likely represents sites of contention, crisis reporting, or political skepticism.

Influencer-Level Sentiment Analysis

To add to the community view, we isolated the top 15 hashtags by Betweenness Centrality and calculated their mean sentiment scores. This approach evaluates whether the network's structural bridges (or "Super Connectors") are characterized by trust-building narratives or critical urgency.

The emotional tone of the network's primary structural bridges reveals distinct emotional patterns rather than neutral tones (Figure 15).

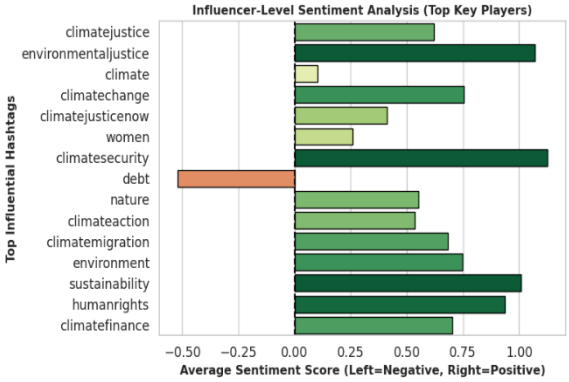

Fig. 15: Influencer Level Sentiment Analysis. The horizontal bar chart ranks the top 15 structural connectors by means sentiment. Green bars indicate constructive/positive framing, while the orange bar represents critical/negative framing.

The results show that #ClimateSecurity holds the highest positive sentiment (1.1247). It is followed by #EnvironmentalJustice (1.0702) and #Sustainability (1.0076). These high positive scores indicate that the hashtags acting as the strongest bridges for information flow are predominantly rooted in solution-oriented, trust-building, and rights-based narratives.

However, a sharp contrast is evident in the structural position of #Debt. Standing as a critical outlier with significant negative sentiment (-0.5199), #Debt acts as a channel for grievance rather than hope. Its negative tone highlights deep seated frustrations regarding climate finance and economic inequality. Meanwhile, broad categorical tags like #Climate (0.1009) and #Women (0.2587) maintain a comparatively lower positive sentiment. This suggests they are often used in more neutral, reporting-based contexts rather than emotive advocacy.

Table 12: Sentiment and Centrality Metrics for Top Structural Bridges

Hashtag	Betweenness Centrality	Frequency	Sentiment Score
climatejustice	0.2337	223,282	0.6195
environmentaljustice	0.1630	47,055	1.0702
climate	0.0414	202,530	0.1009
climatechange	0.0155	202,840	0.7549
climatesecurity	0.0089	7,601	1.1247
climatejusticenow	0.0113	11,660	0.4115
sustainability	0.0021	25,381	1.0076
humanrights	0.0020	9,660	0.9381
women	0.0107	12,779	0.2587
climatemigration	0.0042	4,575	0.6824
nature	0.0055	42,088	0.5508
environment	0.0022	38,010	0.7460
climateaction	0.0047	34,612	0.5374
climatefinance	0.0019	3,424	0.7027
debt	0.0088	4,527	-0.5199

The influencer analysis yields three core insights regarding the network's emotional structure. First, the high positivity of tags like #ClimateSecurity and #Sustainability suggests that "security" concepts are being framed constructively. They facilitate information flow rather than using fear-based narratives. Second, the resilience of #EnvironmentalJustice and #HumanRights confirms that the justice framework serves as a source of moral strength and solidarity. It effectively connects diverse discourse strands. Finally, the negative score of #Debt reveals that economic discussions serve as the primary site of emotional tension. It acts as a negatively charged connector between policy discussions and activist demands. This synthesis confirms that the climate security discourse operates through emotionally differentiated structural pathways. Solution-building flows through positive connectors while contentious financial justice issues flow through critical ones.

7. Conclusion

This research provided a comprehensive analysis of the digital discourse surrounding climate security during the critical period of the COP27 summit. By combining network structure with emotional content analysis, we uncovered a discourse ecosystem that is resilient in some aspects yet fragile in others. It is characterized by several critical contradictions.

The clearest finding is the structural dominance of justice. Hashtags such as #ClimateJustice and #EnvironmentalJustice acted as the primary hubs of

the network. They did not merely represent popular topics but served as essential structural bridges. They linked diverse communities ranging from grassroots activists to global policymakers. This demonstrates that for the online public, the concepts of climate and security are deeply connected to issues of fairness and equity. Justice has become the universal frame that enables distinct perspectives to engage with one another.

However, our analysis identified four defining paradoxes in this digital landscape.

First is the "Dilution Effect," a structural phenomenon where the massive surge in participation during COP27 paradoxically led to a decline in network density. The influx of temporary voices diluted the strong mutual connections maintained by the core activist community. This suggests that global events expand visibility at the cost of community cohesion. This created a state of "diluted stability." Here the summit generated volume without the intense emotional turbulence seen during distinct crisis moments like the September heatwaves.

Second is the "Resilience Paradox," contrary to expectations, the discourse from the Global South regions maintained a significantly more positive and forward-looking tone compared to the Global North. While the North expressed skepticism and anxiety, the South fostered narratives of collective hope and mobilization. This challenges the assumption that vulnerability equates to despair. The region's most vulnerable to climate impacts are actually leading the constructive discourse.

Third is an "Influence Paradox" within the network structure. Our metrics revealed that hashtags with relatively low volume, such as #ClimateScam and #LossAndDamage, occupied strategically powerful positions as hidden bridges. These nodes connected otherwise separate parts of the network. This granted counter narratives and niche policy terms a strong influence on information flow. This confirms that popularity is not synonymous with strategic power.

Finally, a "Structural Thematic Paradox" emerged from our Assortativity analysis. While the network is structurally democratic, meaning hubs connect with diverse users, it remains thematically polarized. Users are connected across the network structure but remain locked within their specific topic bubbles. They are effectively talking at the network rather than with it.

These insights have urgent implications for climate communication. The dominance of the justice frame indicates it must remain the foundation of public messaging to ensure structural connectivity. The regional differences in sentiment highlight the need for specific communication strategies that recognize distinct emotional landscapes. Furthermore, the analysis of structural bridges reveals that the network pathways are emotionally shaped. Solution oriented tags like #ClimateSecurity facilitate positive flows. Meanwhile, financial tags like #Debt act as the primary "friction points" of the network. They function as critical barriers where the conversation consistently shifts from hope to grievance. Ultimately, this study demonstrates that while the digital climate security network is robust enough to survive crises, its potential for unified action is constrained by thematic silos and strategic disinformation pathways.

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